LAB4: Computer Vision

Optical Flow, Object Tracking & Segmentation

Objective:

- *Understand the concept of motion estimation in image sequences using consecutive frames.*
- *Explain the principle of optical flow and visualize pixel-level motion between two images.*
- *Implement optical flow algorithms using OpenCV to detect and analyze object movement with/without using a camera.*

LAB: Computer Vision

Description

- Optical flow finds how pixels move from one image to the next. 📌 We use two consecutive frames instead of a camera.

-Optical Flow Using Two Images

- Example Scenario frame1.jpg → object at position A frame2.jpg → object slightly moved

[1]:
```python
import cv2
import numpy as np
import matplotlib.pyplot as plt

img = cv2.imread("camera.png")

# Create translation matrix (shift right by 10 pixels)
M = np.float32([[1, 0, 10],
                [0, 1, 0]])

shifted = cv2.warpAffine(img, M, (img.shape[1], img.shape[0]))

cv2.imwrite("frame1.jpg", img)
cv2.imwrite("frame2.jpg", shifted)

print("frame1.jpg and frame2.jpg created")
```

```
frame1.jpg and frame2.jpg created
```

- *Optical Flow = how pixels move from one image to the next*

\- Imagine two frames of a video:

Frame 1 → before movement

Frame 2 → after movement

- *Optical flow tells us: How fast a pixel moved In which direction it moved*

[3]:
```python
img1 = cv2.imread("frame1.jpg", 0)
img2 = cv2.imread("frame2.jpg", 0)

flow = cv2.calcOpticalFlowFarneback(
    img1, img2, None,
    0.5, 3, 15, 3, 5, 1.2, 0)

magnitude, angle = cv2.cartToPolar(flow[...,0], flow[...,1])

plt.imshow(magnitude, cmap='gray')
plt.title("Optical Flow Magnitude")
plt.axis('off')
```

[3]:
```
(np.float64(-0.5), np.float64(511.5), np.float64(511.5), np.float64(-0.5))
```

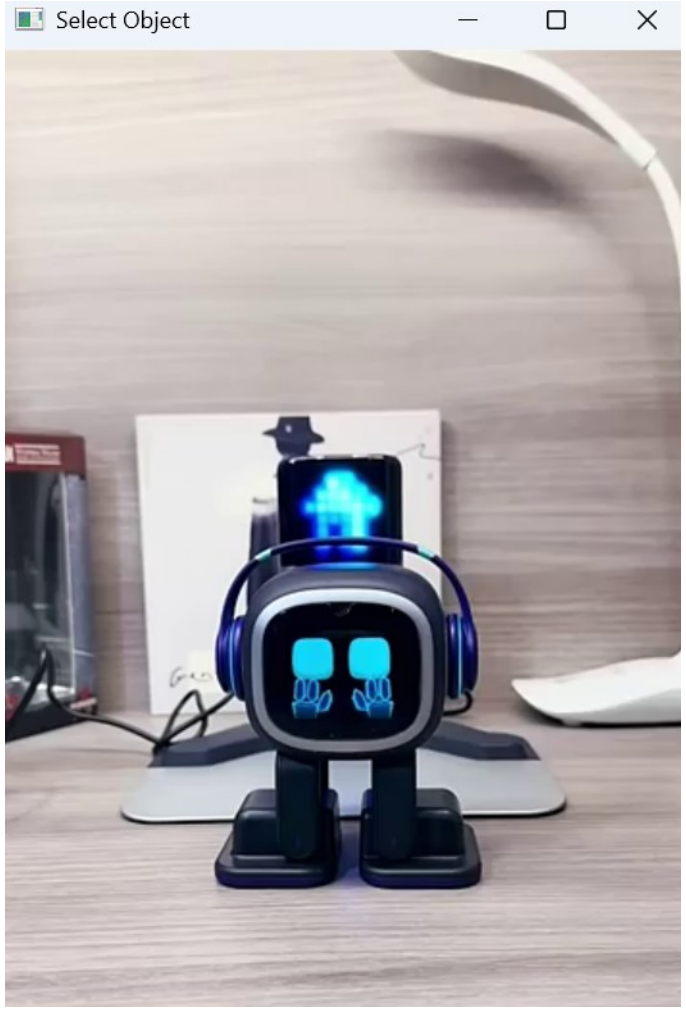
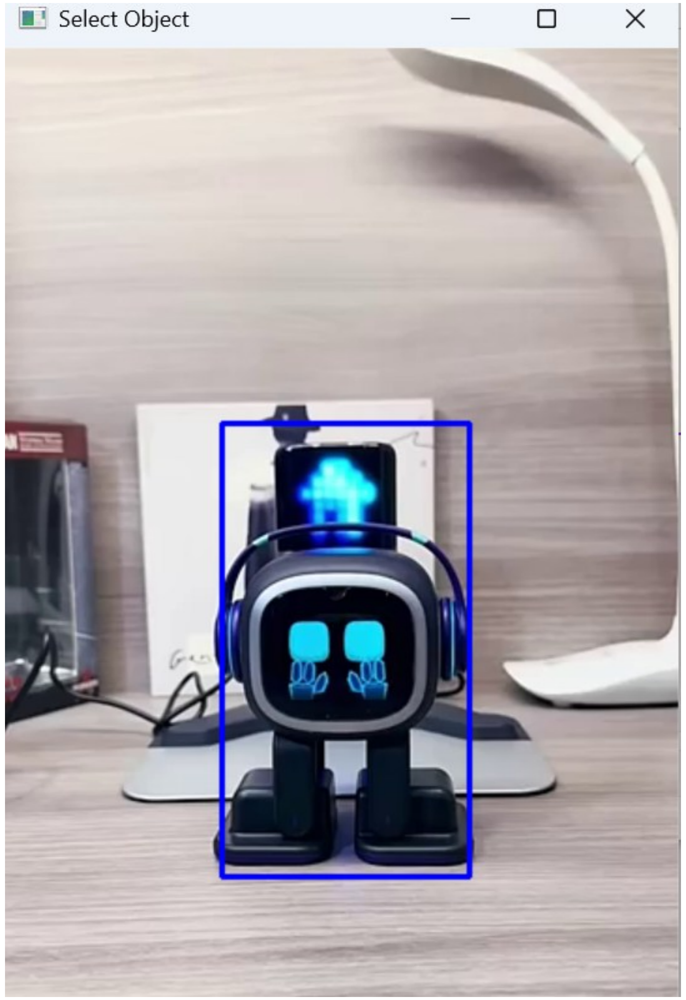
Optical Flow Magnitude

Bright areas → more motion

Dark areas → less motion

[1]:
```python
import cv2

# Load video
cap = cv2.VideoCapture("video1.mp4")

# Read first frame
ret, frame = cap.read()
if not ret:
    print("Cannot read video")
    cap.release()
    exit()

# Create tracker (MOSSE is simplest)
tracker = cv2.legacy.TrackerMOSSE_create()

# Select object to track
bbox = cv2.selectROI("Select Object", frame, False)
tracker.init(frame, bbox)

# Tracking loop
while True:
    ret, frame = cap.read()
    if not ret:
        break

    success, box = tracker.update(frame)
```

```
    if success:
        x, y, w, h = map(int, box)
        cv2.rectangle(frame, (x, y), (x+w, y+h), (0, 255, 0), 2)

    cv2.imshow("Object Tracking", frame)

    if cv2.waitKey(30) & 0xFF == 27:
        break

cap.release()
cv2.destroyAllWindows()
```


Select Object

Select Object

Object Tracking

-Students task:

- **Track a single, clearly visible object (e.g., a ball, box, or moving person) in video1.mp4 using the** (Minimum Output Sum of Squared Error) **MOSSE tracker.**

- *At the end create pdf file and submit, don’t be late, late submission may affect your sessional marks*